Advancing NASA Supply Chain Resiliency

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Supply Chain Risk Management (SCRM)
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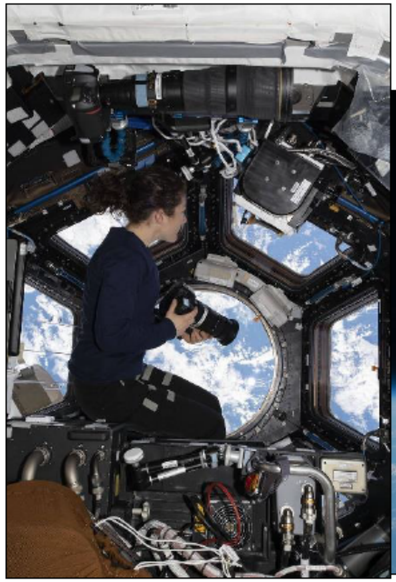
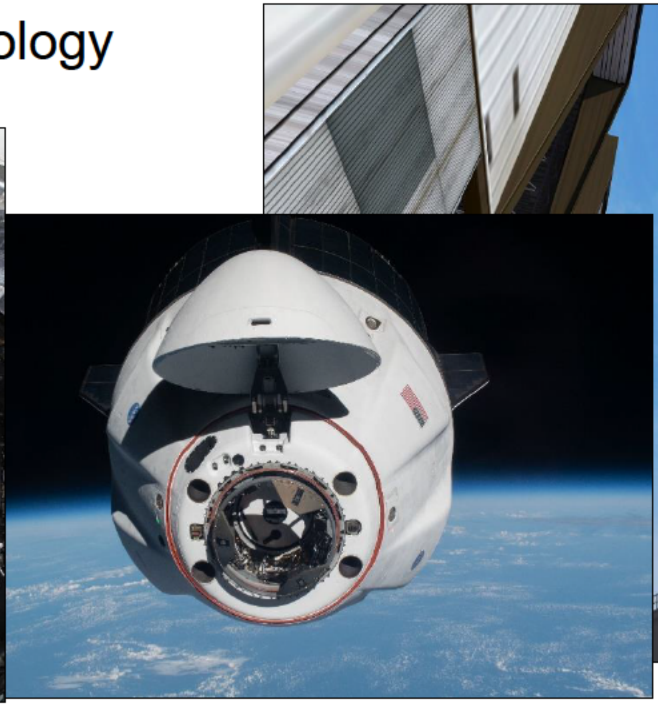
About NASA

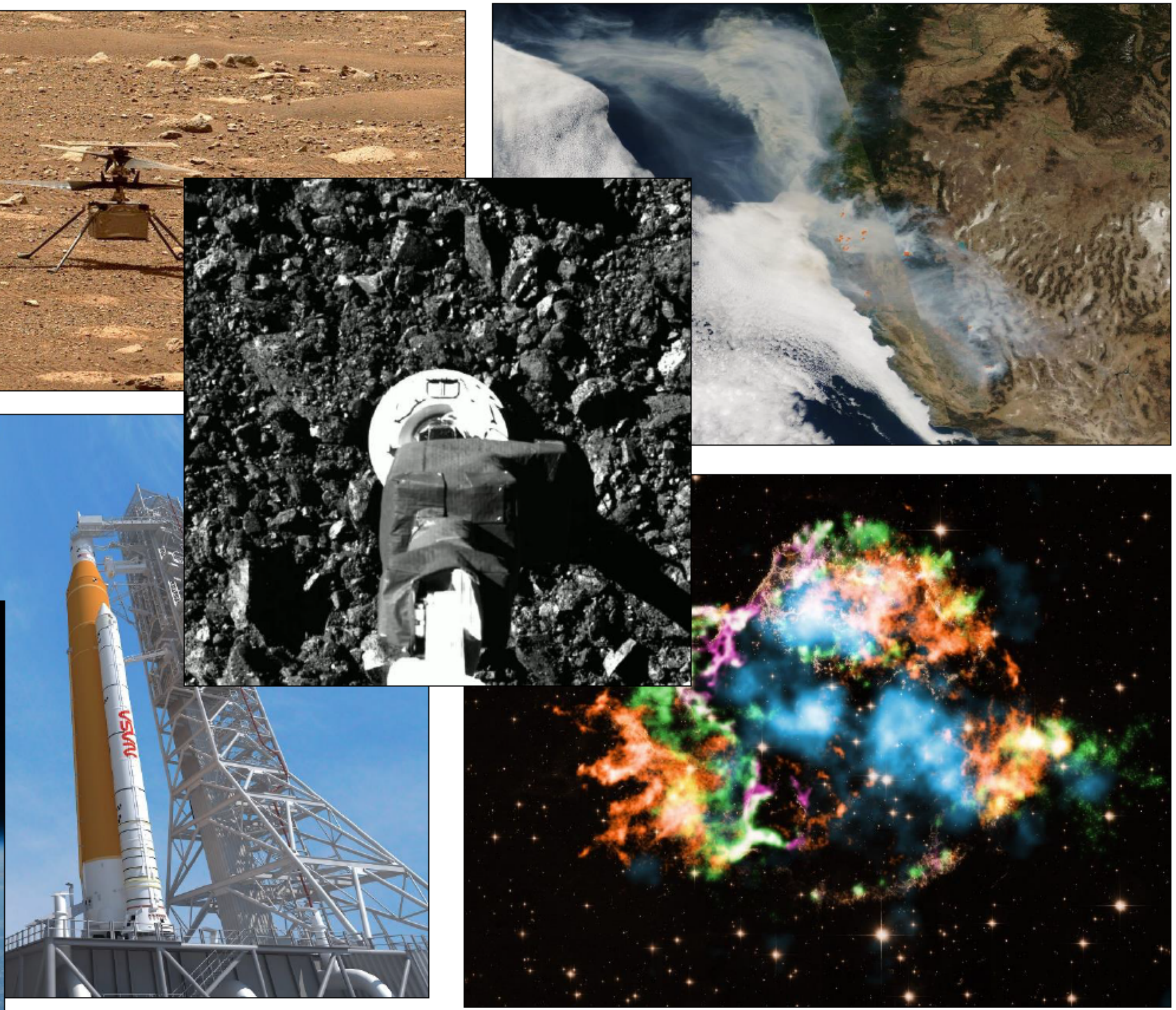
[NASA 2024: Onward and Upward | NASA+]

Mission Directorates

- Aeronautics Research
- Exploration Systems Development
- Space Operations
- Science
- Space Technology







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A World of Risks Threaten Supply Chains

Trends / conditions driving risks include:

- Globalization of markets, technology and industry with supply chains operating across geographical, political, social, cultural and economic environments
  - Supply chains yield advantages and exposure to risks
- Competition and conflicts (e.g., U.S. - China rivalry, Russian invasion of Ukraine)
- Climate change

Dynamic interplay of a wide array of risk types can disrupt supply chains, such as:

Natural Disasters / Hazards

Human-Made Disasters / Hazards

Geopolitical & Economic Events

Regulatory Actions

Litigation

Enterprise Management

Financial Capabilities

Workforce Shortages

Cyberattacks

Disease Pandemics

Reputational / Political

Product / Service Security

Transportation / Logistics

Foreign Influence / Dependency

Diminishing / Sole Sources

Fragile Supplier

New Risk Types: "Unknown Unknowns"

Limited Visibility into Supply Chains = Unknown Risk Exposure

Strategic Situation

- NASA missions rely upon transnational supply chains of commercial, non-profit and government organizations and the U.S. industrial base to develop and operate complex, high-value and innovative systems for the nation
  - Dynamic array of **technical, business, economic, security, and climate risks** threaten to disrupt or deny the **timely, affordable** provisioning of **products and services** as required for mission success
- Administration policy
  - Executive Order 14005, Ensuring the Future is Made in All of America by All of America's Workers *(1/25/2021)*
    - *... the United States Government should ....maximize the use of goods, products and materials produced in, and services offered in, the United States.*
  - Executive Order 14017, America's Supply Chains *(2/24/2021)*
    - *The United States needs resilient, diverse and secure supply chains to ensure our economic prosperity and national security.*
  - United States Space Priorities Framework (12/2021)
    - *.... the United States will ... strengthen the resilience of supply chains across the nation's space industrial base.*
  - Executive Order 14123, White House Council on Supply Chain Resilience (6/21/2024)
    - *The Council shall coordinate and promote Federal Government efforts to strengthen long-term supply chain resilience and American industrial competitiveness ....*

About the OSMA SCRM Program
Launched October 2019

- Program Mission:
  - Advance strategy, policy, processes and capabilities to assure resilient supply chains for NASA mission performance
    - Holistic, multidisciplinary approach to implementing integrated, collaborative and substainable solutions
- Program Goals:
  - Build Supply Chain Visibility and Insight
  - Optimize Government Contractor Quality Assurance
  - Use Established Risk Management Processes
  - Establish SCRM Mindset and State-of-the-Market Capabilities
- Key Initiatives:
  - SCRM Policy, Processes and Partnerships
    - **Agency-level Supply Chain Resiliency Board**
    - **Supply Chain Visibility Reporting** by prime contractors (Data Requirements Description)
    - **Industrial Base and Supply Chain Risk Management (SCRM) Strategy and Status** requirement (NPR 7120.5F / NASA Space Flight Program & Project Management Requirements)
  - Supply Chain Insight Central (SCIC) information and analysis services platform
  - Supply Chain Research & Analysis

Supply Chain Resiliency Board (SCRB)

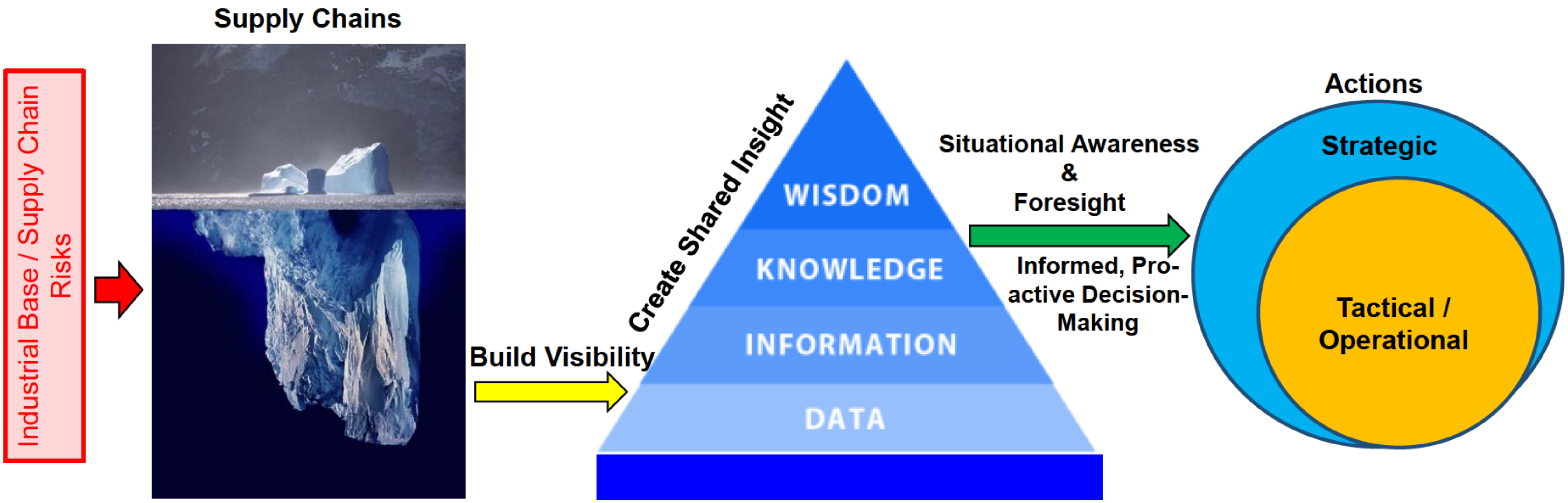
Charter Highlights[1]

- Support the Acquisition Strategy Council (ASC) and the Chief Acquisition Officer (CAO) in advancing Agency strategy, policy, processes, capabilities, and organizational culture for (1) the pro-active assessment and management of supply chain and industrial base risks and opportunities to assure resilient NASA mission performance, and (2) the fulfillment of applicable U.S. Government policy and statutory requirements.
- Assess, formulate, and recommend courses of action and solutions for review and decision-making by member organizations, the ASC, the ASC Chair and/or the CAO.
- Assess, submit, and report on risks and associated mitigations as part of the NASA Enterprise Risk Management process.
- Support NASA-level coordination, review and decision-making pertaining to NASA strategy, policy, initiatives, and actions, including matters involving legislation, public laws, executive orders, and strategic stakeholders.
- Make decisions on board recommendations, actions, and activities.

Note: (1) Go to https://nodis3.gsfc.nasa.gov/OPD_Docs/NC_1000_56_.pdf for the charter publicly released on 11/02/2023

Supply Chain Insight Central Information & Analysis Services

Strategy - Operational Since March 2021



Enterprise SCRM Risk: limited visibility and insight into supply chains = unknown risk exposure and insufficient information for analysis, planning and management

Key Strategy: systematically bring data and information together to enable pro-active, collaborative SCRM across the agency

Supply Chain Visibility (SCV) Reporting Implementation

Effective Agencywide **6/28/2024**

Supply Chain Visibility (SCV) Reporting

Supply Chain Visibility Reporting Requirements

EFFECTIVE DATE: 06/28/2024

Welcome to NASA's internal site for Supply Chain Visibility (SCV) Reporting Requirements!

NASA mission programs and projects rely upon the U.S. industrial base and transnational supply chains of commercial, non-profit and government organizations to develop and operate complex, high-value, and innovative systems for the nation. NASA prime contractors and their sub-tier suppliers are subject to a dynamic array of risks which threaten the timely, affordable provisioning of products and services as required for mission success. Accordingly, as part of ongoing efforts to strengthen agency capabilities to pro-actively recognize, assess and address industrial base and supply chain risks and opportunities the NASA Supply Chain Resiliency Board (SCRB) authorized in November 2023 the implementation of the baseline Data Requirements Description (DRD) for SCV Reporting. The effective date for the agencywide implementation of the SCV Reporting DRD is June 28, 2024.

The purpose of this site is to provide the current SCV Reporting requirements along with supporting guidance and assistance to implement and monitor the reporting and produce the visibility key to assuring the resiliency of supply chains for NASA mission performance.

The Supply Chain Risk Management (SCRM) program within the NASA Office Safety and Mission

Key Links
Procurement Information Circular (PIC) 24-04: Implementation of SCV Reporting
2020 NASA Technology Taxonomy
Agency Mission Program & Project List
NASA Form 1634 Contracting Officer's Representative - COR / Alternate COR Delegation
NASA Form 1707 Special Approvals and Affirmations of Requisitions
NASA Supply Chain Resiliency Board Charter
Supply Chain Insight Central (SCIC) Information and Analysis Services
Supply Chain Visibility (SCV) Reporting -Prime Contractor User Registration

NASA Office of Procurement

Procurement Dispatch Instant Alert
June 28, 2024
Policy

Supply Chain Visibility (SCV) Reporting Implementation Notice

In accordance with the decision of the NASA Supply Chain Resiliency Board (SCRB) to implement Supply Chain Visibility (SCV) Reporting by NASA prime contractors as a key part of agency efforts to pro-actively address U.S. industrial base and supply c…

National Aeronautics and Space Administration
Washington, DC 20546

Procurement Information Circular

PIC 24-04

July 16, 2024

IMPLEMENTATION OF SUPPLY CHAIN VISIBILITY (SCV) REPORTING (NFS Case 2024-N014)

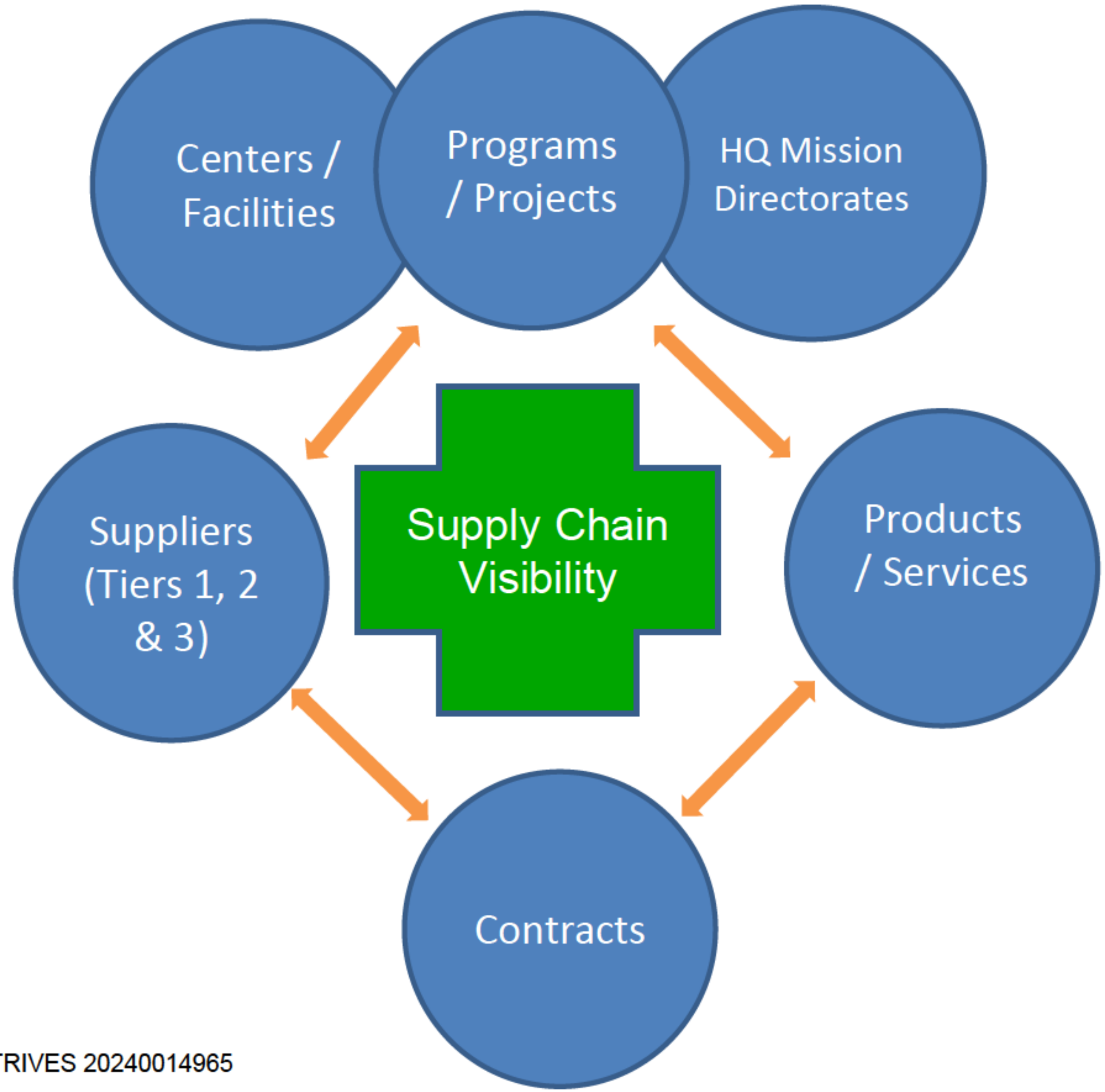
PURPOSE: Provide instructions to Contracting Officers to implement the approved baseline Data Requirements Description (DRD) for Supply Chain Visibility (SCV) Reporting by NASA prime contractors into NASA contract solicitations and awards for new procurements of products and services for programs / projects on the approved Agency Mission Program and Project List (AMPL) with estimated contract values of $20 million or more. Requiring organization: Office of Safety and Mission Assurance (OSMA) on behalf of the NASA Supply Chain Resiliency Board.

BACKGROUND: NASA mission programs and projects rely upon the U.S. industrial base and transnational supply chains of commercial, non-profit and government organizations to develop and operate complex, high-value, and innovative systems for the nation. NASA prime contractors and their sub-tier suppliers are subject to a dynamic array of risks which threaten the timely, affordable provisioning of products and services as required for NASA mission success. As part of ongoing efforts to strengthen agency capabilities to proactively recognize, assess and address industrial base and supply chain risks and opportunities the NASA Supply Chain Resiliency Board (SCRB) decided on November 30, 2023 to proceed with the implementation of Supply Chain Visibility (SCV) Reporting by NASA prime contractors. SCV has been coordinated with all Mission Directorates, Agency Leadership and other stakeholders.

The following items are implemented and effective as of June 28, 2024 to support and assure the incorporation of SCV Reporting requirements into NASA procurements as applicable:

Supply Chain Visibility (SCV) Reporting

- Applicable to $20m+ NASA procurements of products and services for programs / projects on the Agency Mission Program and Project List
- Limited to specific information on supplier entities within the top three contractual tiers
  - Tier 3: direct suppliers → Tier 2: direct sub-contractors →Tier 1: prime contractors to NASA with tailoring for deeper or more frequent reporting than baseline requirements
- Reporting via NASA's online, secure Supply Chain Insight Central (SCIC) information system to enable integrated data/information management for internal NASA analysis, planning and decision-making at project to enterprise levels



Industrial Base / SCRM Strategy

Project Life-Cycle

NPR 7120.5F (NASA Space Flight Program & Project Management Requirements, effective 8/3/2021) in combination with NPR 8735.2C (Hardware Quality Assurance Program Requirements for Programs & Projects) requires ***Industrial Base (IB) and Supply Chain Risk Management (SCRM) Strategy and Status*** for Mission Programs and Project

Acquisition Strategy Meeting

Assess key Industrial Base / SCRM factors

System Requirements & Definition Reviews

Preliminary Project Plan

- Industrial Base / SCRM Strategy

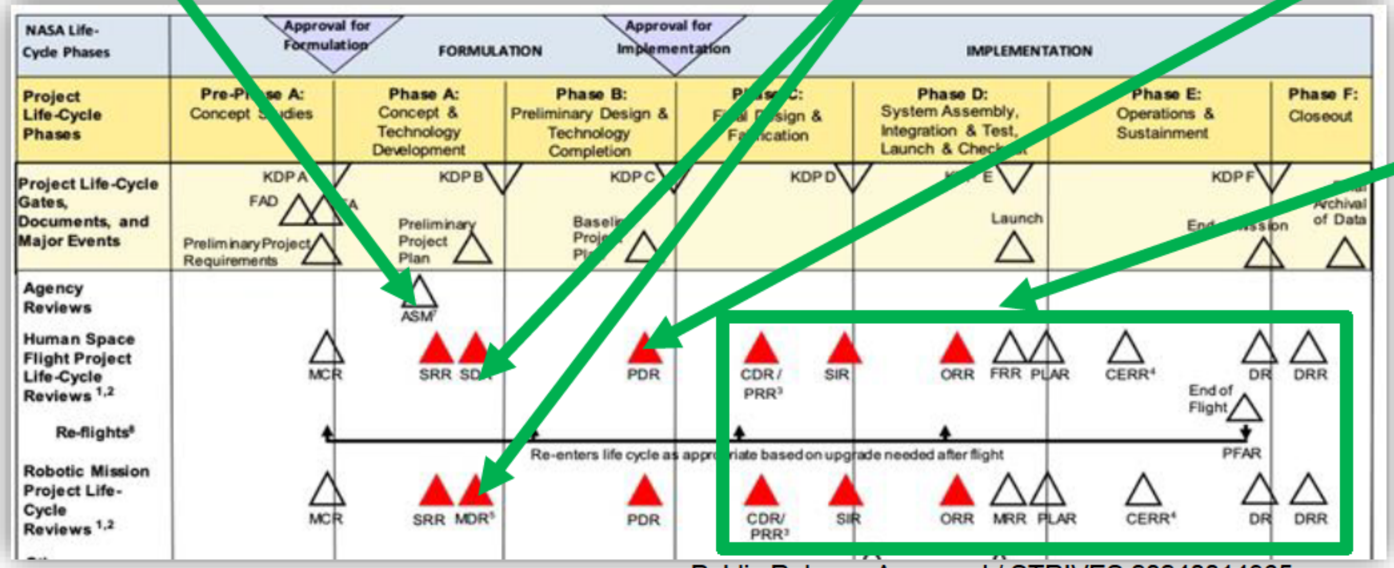
Supply Chain Visibility Reporting by Prime Contractors

Baseline Project Plan

- Industrial Base / SCRM Strategy

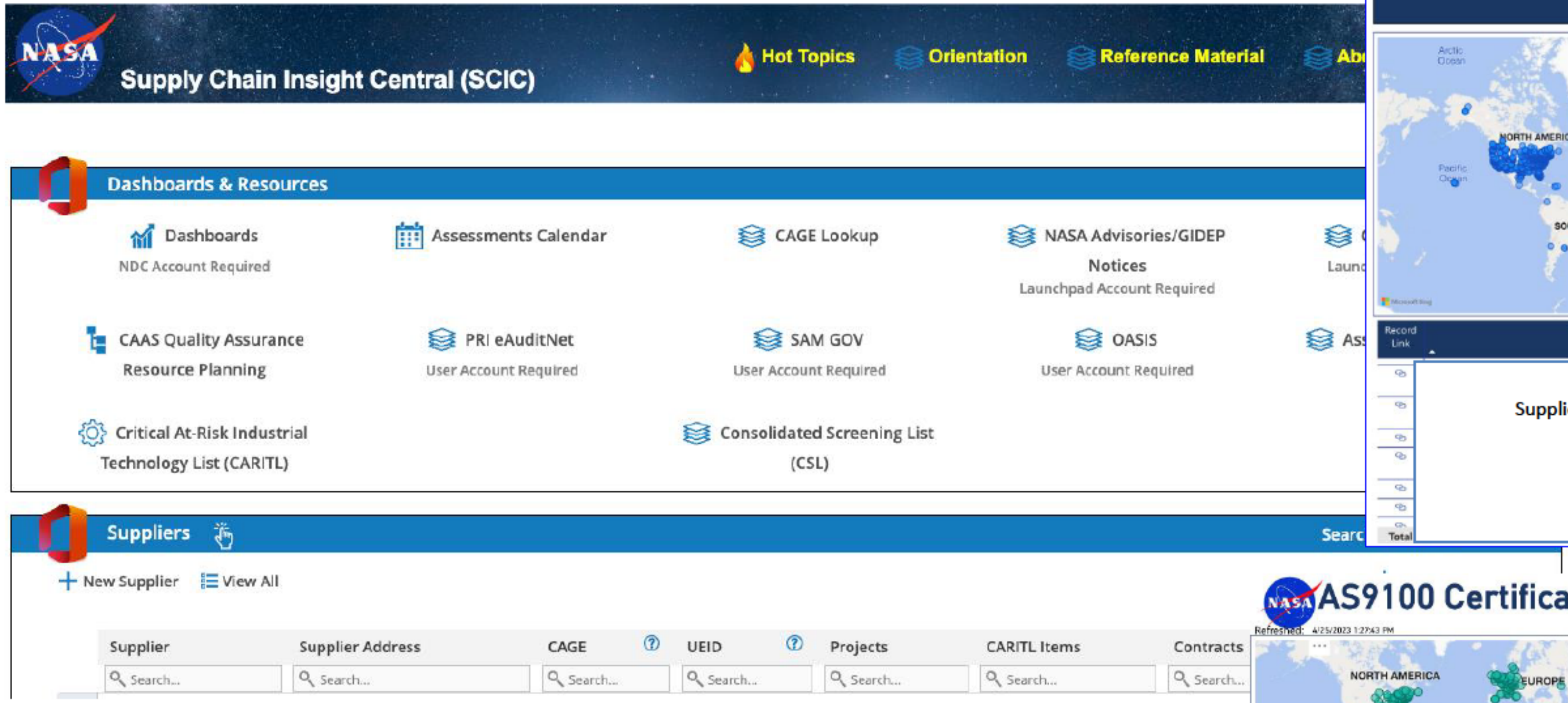
Supply Chain Visibility Reporting by Prime Contractors

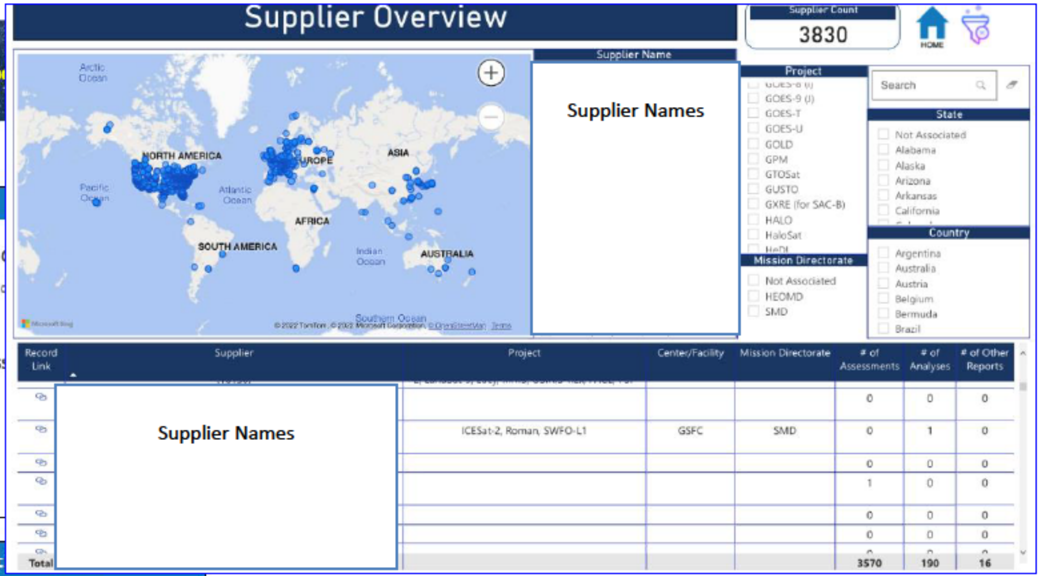
Ongoing Industrial Base / SCRM Strategy Implementation & Supply Chain Visibility Reporting

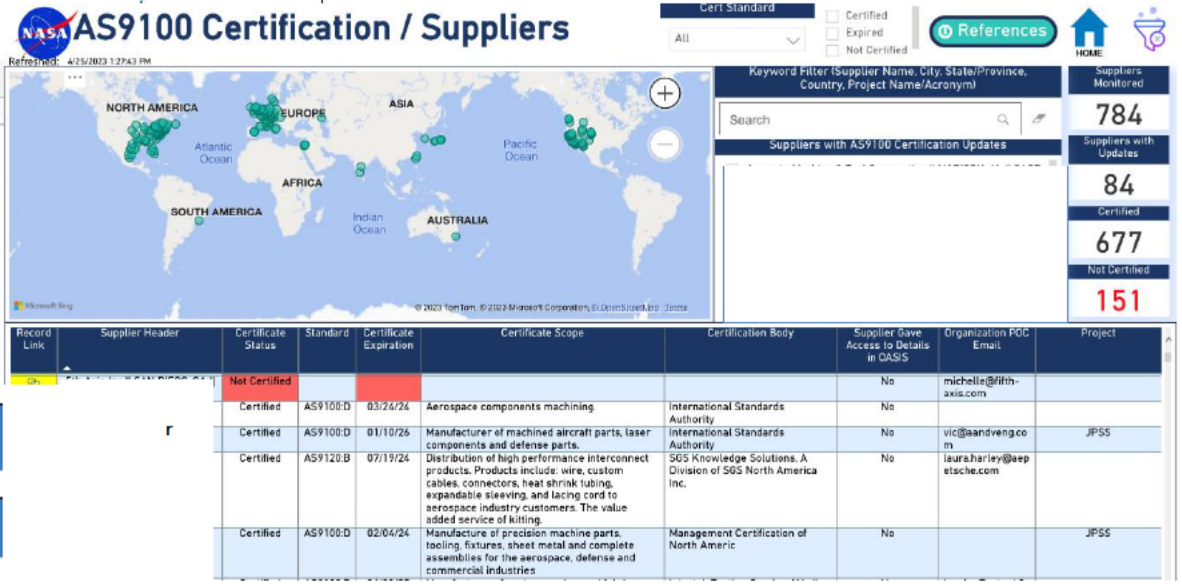


SCIC Information & Analysis Services

Operational Since March 2021







Suppliers

Contracts

Supplier Assessments

Products & Services

Supplier Research & Analysis (SRA) Reports

Projects

Other Supplier Reports

Dashboards & Resources

Supply Chain Research & Analysis

Purpose:

- Build visibility and insight into suppliers / supply chains to enable pro-active SCRM

Approach:

- Holistic analytical framework, business intelligence techniques, robust information sources,
- Non-intrusive to suppliers; internal NASA/USG use only
- Right-sized: profiles → briefings → in-depth reports

Analytics / Dashboards:

- Interactive visual displays with integrated data

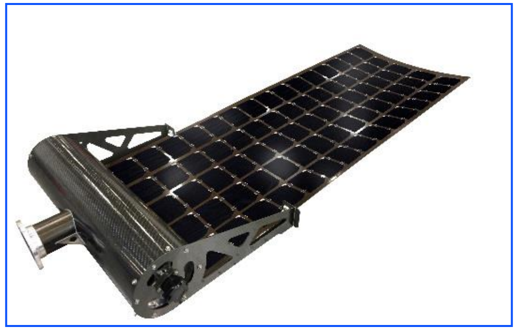
Financial Capability Assessments:

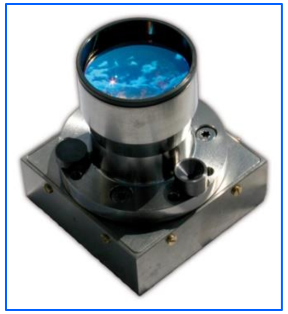
- Produce assessments for $500m+ procurements per NASA FAR 1809.105-1 and for other procurements as requested

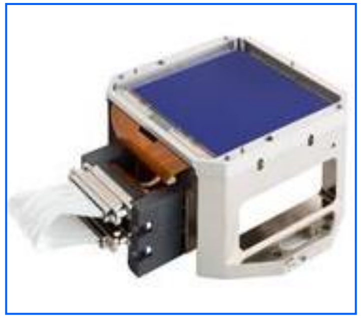
U.S. Civil Space Industrial Base Survey:

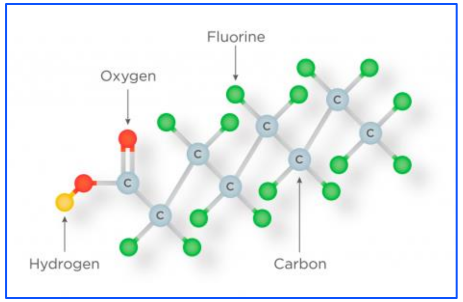
- Teamed with HQ/Office of Technology, Policy and Strategy and DOC's Bureau of Industry & Security to manage and analyze survey data

Some Products of interest:


Solar Array

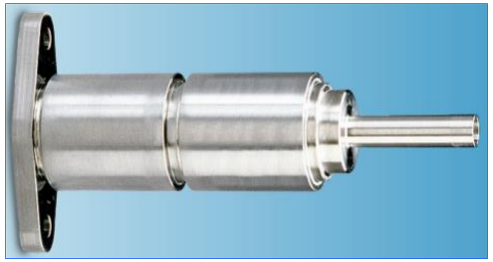

Star Tracker


Charge-coupled device



PFAS Chemicals


Lithium-Ion Battery


Bipropellant Thruster Valve

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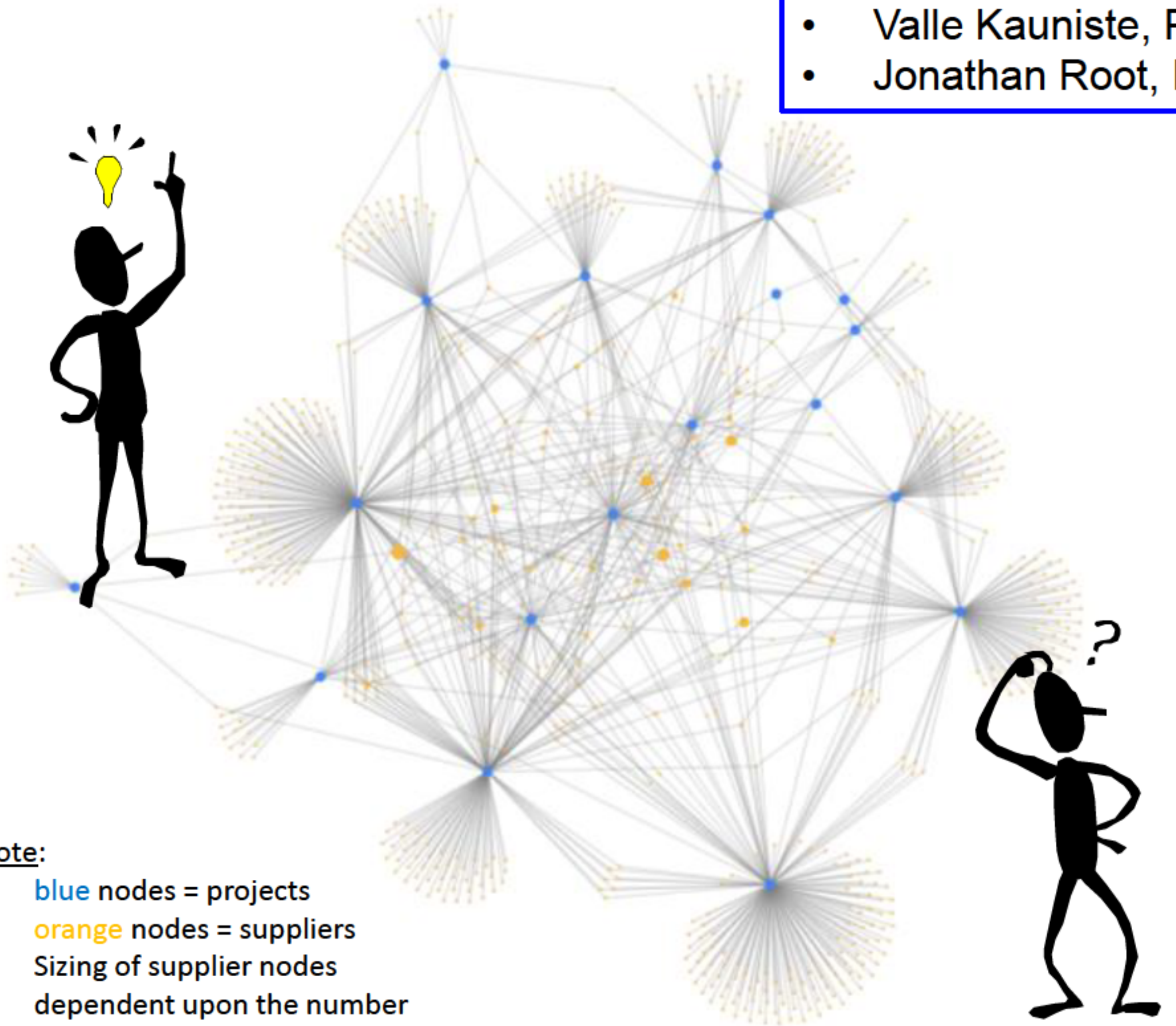
Discussion

Ideas, Opportunities or Questions? Contact the OSMA SCRM program:

- Valle Kauniste, Program Executive (valle.j.kauniste@nasa.gov)
- Jonathan Root, Deputy Program Executive (jonathan.f.root@nasa.gov)



Note:
- blue nodes = projects
- orange nodes = suppliers
- Sizing of supplier nodes dependent upon the number of project relationships

An Old Proverb

For want of a nail the shoe was lost;
For want of a shoe the horse was lost;
For want of a horse the rider was lost;
For want of a rider the battle was lost;
For want of a battle the kingdom was lost;
And all for the want of a horseshoe nail.

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